

1. Cover Page

- Project title
 - Course name
 - Group number
 - Student names + roll numbers
 - Instructors name
 - Institution
 - Submission date
-

2. Abstract

Brief summary:

- problem statement
- solution overview
- tools used
- outcome

Should be readable independently.

3. Introduction

Explain:

- background of the problem
 - real-world relevance
 - objectives
-

4. Problem Statement

Clear definition of:

What problem are we solving?

5. System Overview

High-level architecture.

Include:

- block diagram
- system workflow
- major modules

Explain how components interact.

6. Hardware Description

List all components:

- Arduino board
- sensors
- actuators
- modules

For each component:

- function

- working principle
 - reason for selection
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7. Software Design

Explain:

- algorithm
- flowchart
- logic design
- control strategy

Include:

- Arduino IDE details
 - libraries used
 - code structure
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8. Circuit Diagram

- schematic diagram
 - wiring explanation
 - pin configuration table
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9. Implementation

Describe actual building process:

- assembly
- testing stages
- integration

Include photos of prototype.

10. Results and Observations

- experimental outputs
- measurements
- screenshots
- behaviour analysis

Explain what worked and what didn't.

11. Challenges Faced

Discuss:

- hardware issues
- sensor noise
- debugging problems
- time constraints

Shows learning process.

12. Applications

Real-world uses of the system.

Example:

- smart homes
- industry
- healthcare
- safety

13. Future Improvements

Ideas for expansion:

- IoT integration
- AI
- mobile app
- cloud analytics

14. Conclusion

Summarize:

- achievements
- learning outcomes
- project success